

①. Are these system specifications consistent?

A: Whenever the system software is being upgraded, users cannot access the file system. $\hookrightarrow p$

B: If users can access the file system, then they can save new files. r

C: If users cannot save new files, then the system software is not being upgraded.

Sol.
=

$$A : p \rightarrow \neg q$$

$$B : q \rightarrow r$$

$$C : \neg r \rightarrow \neg p$$

(A)

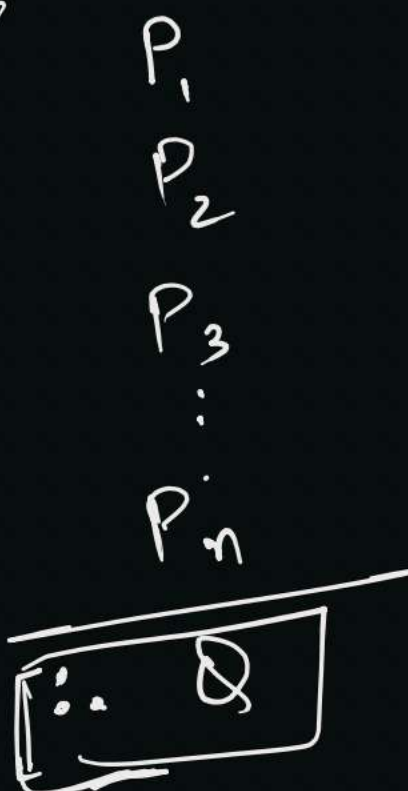
(B)

(C)

$$C : \neg r \rightarrow \neg p$$

p	(2)	(r)	$\neg p$	$\neg r$	$\neg \neg r$	(A) $p \rightarrow \neg r$	(B) $r \rightarrow r$	(C) $\neg r \rightarrow \neg p$
T	T	T	F	F	F	F $\times$	—	—
T	T	F	F	F	T	F $\times$	—	—
T	F	T	F	T	F	T	T	T =
T	F	F	F	T	T	T	T	—
F	T	T	T	F	F	T	F $\times$	—
F	T	F	T	F	T	T	T	—
F	F	T	T	T	F	T	T	—
F	F	F	T	T	T	T	T	—

argument form



yes/no?

$(P_1 \wedge P_2 \dots \wedge P_n) \rightarrow Q \rightarrow$ it should be a tautology.

$\hookrightarrow$ sound valid arguments.

$\hookrightarrow$ unsound ones. $\boxed{f \rightarrow f/T} \Rightarrow \text{valid}$

valid arguments are always consistent. $\checkmark$

$(A \rightarrow B) \rightarrow \text{true}$
 $\begin{matrix} F & T/F \end{matrix}$

- (2.) A: If the file system is not locked, then new messages will be queued.
- (H.W.) B: If the file system is not locked, then the system is functioning normally, and conversely.
- C: If new messages are not queued, then they will be sent to the message buffer.
- D: If the file system is not locked, then new messages will be sent to the message buffer.
- E: New messages will not be sent to the message buffer.

Consistent or not?

③ Are they logically equivalent!

a) $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ $\neq$

H.W. b) $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ $\neq$

Sol.

$\checkmark$ p	q	r	$\textcircled{x}$ $p \rightarrow q$	$\checkmark$ $\textcircled{y}$ $(q \rightarrow r)$	$x \rightarrow r$	$p \rightarrow y$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

$\neq$ not equivalent.

(4.) Find the dual of each of these compound propositions.

a) $\boxed{p \vee \neg p} = S \Rightarrow \text{Dual}(S) = S^*$

b) $p \wedge (q \vee (r \wedge T))$

c) $(p \wedge \neg q) \vee (q \wedge F)$

Any compound proposition that contains $\wedge$, $\vee$ and $\neg$, then its dual will be :

Just replace them $\left\{ \begin{array}{ll} \wedge \rightarrow \vee \\ \vee \rightarrow \wedge \\ T \rightarrow F \\ F \rightarrow T \end{array} \right.$

Sol. a) dual of $(p \vee \neg p) \Rightarrow p \wedge \neg p$

b) $p \vee (q \wedge (r \vee F))$

c) $(p \vee \neg q) \wedge (q \vee T)$

Q. Does $S^* = S$? no.

* Does $(S^*)^* = S$? Yes, $S \rightarrow$ compound proposition.

5. Determine whether these arguments are valid or not.

valid. a) Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore Mia is not enrolled in the university. } modus tollens.

Invalid. b) Quincy likes all action movies. Quincy likes the movie Eight Men Out. Therefore, Eight Men Out is an action movie. } fallacies.

Invalid. c) A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.

⑥ a) If x is a positive real number, then x^2 is a positive
Invalid real number. Therefore, if a^2 is positive, where a is
 a real number, then a is a positive real number.

b) If $x^2 \neq 0$, where x is a real number, then $x \neq 0$.
Valid. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.
 $\therefore$

Determine whether these arguments are valid or not.

Solⁿ. a) $a = -5$, $\boxed{a^2 = 25}$ $\therefore$

b) $p \rightarrow q$

$\frac{p}{\therefore q}$

modus
ponens.